

# vasic-digital tmux — CLAUDE.md

## INHERITED FROM constitution/CLAUDE.md

All rules in constitution/CLAUDE.md and the constitution/Constitution.md it references apply **unconditionally** to this project. Project-specific rules below **extend** them — they do NOT weaken or override any universal clause. When this file disagrees with the constitution submodule, **the constitution wins**.

@constitution/CLAUDE.md

## MANDATORY ANTI-BLUFF END-USER-QUALITY COVENANT

**Forensic anchor — verbatim user mandate (2026-04-28, reasserted 2026-05-21):**

"We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This **MUST NOT** be the case and execution of tests and Challenges **MUST** guarantee the quality, the completion and full usability by end users of the product!"

**Operative rule.** The bar for shipping is **not** "tests pass" but **"end users can actually use the feature."** Every PASS — test OR HelixQA Challenge — **MUST** carry positive captured runtime evidence that the feature works for the end user. Metadata-only PASS, configuration- only PASS, "absence-of-error" PASS, and grep-without-runtime PASS are all critical defects regardless of how green the summary line looks. Tests and Challenges are bound **EQUALLY**.

This covenant is restated verbatim in every governance file at the consumer layer (CLAUDE.md, AGENTS.md, QWEN.md) so any tool that does not expand @imports still reads it. The same verbatim block lives upstream at constitution/Constitution.md §11.4, constitution/CLAUDE.md, and constitution/AGENTS.md. Removing or weakening this block is a §11.4-class release blocker.

Canonical authority: constitution/Constitution.md §11.4 and its sub-anchors §11.4.1 through §11.4.78. Project anchor: Constitution.md §101.

Non-compliance is a release blocker regardless of context.

**Read first on a fresh conversation:** CONTINUATION.md (§0/§3/§8 are mandatory), then Issues.md for OPEN/PARTIAL/BLOCKED/RUNNING items. Canonical project authority: Constitution.md (Project Articles §101–§109) which extends constitution/Constitution.md.

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## Project overview

Verified hardened tmux 3.6a build with jemalloc, OOM protection, and per-session isolation via the tmx wrapper. Built around a hard verification gate that refuses to expose the binary unless functional tests pass. **Native dual-OS** (since 2026-05-13): runs as a host process on Linux AND macOS — no VM in the daily-use path. Each `tmx new -s NAME` creates its own tmux server with OS-native isolation: cgroup-v2 transient scope (`systemd-run --user --scope`) on Linux; POSIX rlimit wrapper (`RLIMIT_CPU + RLIMIT_NPROC`) on macOS. The session shell is the operator's host shell with full `$PATH`. Honest gap: `RLIMIT_AS` (memory) is NOT enforced by XNU for unprivileged processes — see `docs/guide/README.md §5.6`.

## Critical base rules restated (for sessions that don't expand `@imports`)

- **No bluffing.** Every PASS — test OR HelixQA Challenge — carries positive runtime evidence (cgroup/`/proc` file *content*, `capture-pane` output, kernel log lines), never just an exit code. Universal §11.4 / Project §101.
- **FAIL-bluffs equally forbidden.** A test that exits FAIL for a script bug (not a product defect) is fixed at the source layer, never at call sites. Universal §11.4.1.
- **No-guessing.** Never use likely/probably/maybe/seems/appears in cause descriptions — prove with forensic evidence or mark `UNCONFIRMED: / PENDING_FORENSICS:`. Universal §11.4.6.
- **Operator-path coverage.** Every gate test exercises the SAME entry point an end-user invokes (`tmx new -s X`, not hand-spawned equivalents). Project §102.
- **Four-layer coverage.** Every defect lands all four layers before the cycle closes. Project §103.
- **CONTINUATION.md** updated in the SAME commit as any non-trivial state change. Universal §12.10.
- **commit\_all.sh only.** Never `git push` directly on the main repo.

## Commands (exact)

| Step                                                             | Command                                                       |
|------------------------------------------------------------------|---------------------------------------------------------------|
| Install build deps (one-time; Linux needs root, macOS uses brew) | <code>bash scripts/install_deps.sh</code>                     |
| Full pipeline                                                    | <code>bash scripts/setup.sh</code>                            |
| Build only                                                       | <code>bash scripts/setup.sh --build-only</code>               |
| Verify only                                                      | <code>bash scripts/setup.sh --verify-only</code>              |
| Verification gate                                                | <code>bash scripts/verify.sh</code>                           |
| Run all tests                                                    | <code>bash scripts/tests/run_all.sh</code>                    |
| Single test                                                      | <code>bash scripts/tests/NN_*.sh</code> (zero-padded, 01..17) |

| Step                                    | Command                                                                                                                                 |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Constitution inheritance gate           | <code>bash scripts/tests/test_constitution_inheritance.sh</code>                                                                        |
| Meta-test (paired mutation)             | <code>bash scripts/tests/meta_test_false_positive_proof.sh</code>                                                                       |
| Containerized test run (bounded subset) | <code>bash scripts/test_containerized.sh</code>                                                                                         |
| VM test run (full suite)                | <code>bash scripts/test_vm.sh</code><br>( <code>TMX_TEST_DESTRUCTIVE=1</code> for tests 12/13/14;<br><code>META=1</code> for meta-test) |
| End-to-end automation                   | <code>bash scripts/test_e2e.sh</code>                                                                                                   |
| Commit + push (github+gitlab)           | <code>bash commit_all.sh "message"</code>                                                                                               |
| Per-session wrapper                     | <code>tmx {new attach ls kill}</code>                                                                                                   |

Never `git push` directly — use `commit_all.sh`.

## Structure

| Path                                                         | Role                                                                                                                                        |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <code>constitution/</code>                                   | HelixConstitution submodule — universal governance (pinned 7f738df) — <b>do not modify</b>                                                  |
| <code>tmux/</code>                                           | upstream submodule (tag 3.6a) — <b>do not modify</b>                                                                                        |
| <code>Containers/</code>                                     | vasic-digital cgroup helpers submodule                                                                                                      |
| <code>tmux/build*/</code>                                    | build output ( <code>.gitignore'd</code> )                                                                                                  |
| <code>scripts/</code>                                        | build, verify, install, 17 tests + inheritance gate, challenges, wrapper + conf templates                                                   |
| <code>scripts/tmux</code>                                    | generated dispatcher ( <code>.gitignore'd</code> ) — from <code>tmx.template</code> (Linux) or <code>tmx-mac.template</code> (macOS bridge) |
| <code>scripts/tmux.conf.template</code>                      | the tmux config template (scrollback, copy-mode, Claude-Code TUI passthrough, clipboard)                                                    |
| <code>scripts/tests/meta_test_false_positive_proof.sh</code> | §103 layer-4 paired-mutation harness                                                                                                        |
| <code>scripts/challenges/tmux.yaml</code>                    | HelixQA Challenge specs                                                                                                                     |
| <code>commit_all.sh</code>                                   | only allowed push mechanism                                                                                                                 |
| <code>Constitution.md</code>                                 | project authority — Project Articles §101–§109 (extends <code>constitution/</code> )                                                        |

| Path            | Role                                          |
|-----------------|-----------------------------------------------|
| Issues.md       | OPEN / PARTIAL / BLOCKED / RUNNING only       |
| Fixed.md        | RESOLVED items with closure<br>SHA + evidence |
| CONTINUATION.md | live handoff state (must stay fresh)          |

## Project-specific MANDATORY constraints

### Build / packaging

- `scripts/setup.sh` detects host OS and invokes the right build pipeline (Linux ELF / macOS Mach-O) — Project §108.
- `scripts/tmux.conf.template` is the source for the generated tmux config; `setup.sh` substitutes OS-specific placeholders (clipboard command). Never hand-edit the generated `~/tmux.conf` / wrapper.

### Test / verification

- **Test-interrupt-on-discovery** (§103): any defect found during a test cycle STOPS the cycle. Fix at root cause + land all four layers (pre-build gate / runtime test / HelixQA Challenge / paired mutation)
  - rebuild + repeat from the beginning.
- **Operator-path coverage** (§102): gate tests use `tmx new -s X`.
- **Topology dispatch** (§104): tests detect topology and SKIP-with- reason — never silently degrade.
- **Destructive tests** (tests 12/13/14): require `TMX_TEST_DESTRUCTIVE=1` — run only on dedicated test hosts.
- **Constitution inheritance** is itself gated: `scripts/tests/test_constitution_inheritance.sh` + the CM-CONSTITUTION-INHERITANCE paired mutation.

### Deployment / release

- `bash commit_all.sh "<msg>"` is the ONLY push mechanism — pushes to github + gitlab.
- `VERSION` is the single source of truth (`version= semver + versionCode= monotonic`). Both bump together with a `CHANGELOG.md` entry on every release.

## Project-specific Applied Fixes Reference

| #   | Fix                                                    | Key files                               |
|-----|--------------------------------------------------------|-----------------------------------------|
| A13 | Per-session cgroup isolation (OOM contained per scope) | <code>scripts/tmx.template</code>       |
| A15 |                                                        | <code>scripts/tmux.conf.template</code> |

| #   | Fix                                                                 | Key files                                                         |
|-----|---------------------------------------------------------------------|-------------------------------------------------------------------|
|     | Status bar showed<br>claude.exe — strip<br>.exe from window name    |                                                                   |
| A16 | tmux scrolling fix for<br>Claude Code TUI +<br>mobile/Termux        | scripts/tmux.conf.template, scripts/<br>tests/17_*.sh             |
| A17 | HelixConstitution<br>submodule + verified<br>governance inheritance | constitution/, scripts/tests/<br>test_constitution_inheritance.sh |

Full forensic detail: Fixed.md.

## Files to never edit directly

- scripts/tmx — generated by setup.sh from scripts/tmx.template
- tmux/build\*/ — build output (in .gitignore)
- tmux/ — upstream submodule pinned to tag 3.6a
- constitution/ — HelixConstitution submodule (governance authority)

## Project overrides of universal rules

None. This project introduces no override of any universal clause.

## §11.4.87–98 — newly-inherited anchors (constitution 6828ff2)

The constitution submodule was ff-merged to 6828ff2 (2026-05-28). Twelve new universal anchors §11.4.87 through §11.4.98 apply unconditionally per §11.4.35. Long-form bodies live in constitution/Constitution.md; the short-form covenant mirrors below are operative.

### §11.4.87 — Endless-loop autonomous work + zero-idle agent dispatch

Inherited from constitution/Constitution.md §11.4.87. Agent work continues until ALL of: Issues.md has zero non-terminal entries, CONTINUATION.md §3 active-work empty, no in-flight subagent, no external dependency in-flight. Every closure carries §11.4.4(b) four-layer coverage + captured-evidence per §11.4.5/ §11.4.69. Tests AND HelixQA Challenges bound equally per the §11.4 forensic anchor. **Canonical authority:** constitution/Constitution.md §11.4.87.

### §11.4.88 — Background-push (commit-flock released immediately, push detached)

Inherited from constitution/Constitution.md §11.4.88. `commit_all.sh` MUST release the commit flock as soon as `git commit` returns 0; the push runs detached via `nohup ... & + disown`. Per-remote `.git/.push.<remote>.lock` serialises same-remote concurrent pushes but allows different-remote parallelism. The only escape hatch is `--sync-push` for §11.4.41 force-push paths. **Canonical authority:** constitution/Constitution.md §11.4.88.

### §11.4.89 — Background test execution

Inherited from constitution/Constitution.md §11.4.89. Any test expected to exceed ~30 s (`run_all.sh`, `meta_test_false_positive_proof.sh`, `test_e2e.sh`, etc.) MUST spawn via `nohup ... > <log> 2>&1 & + disown`; main work stream returns to the §11.4.42 priority queue immediately. Hard-dependent next steps poll the exit-status file. **Canonical authority:** constitution/Constitution.md §11.4.89.

### §11.4.90 — Obsolete status + per-item obsolescence audit

Inherited from constitution/Constitution.md §11.4.90. The §11.4.15 Status closed-set gains terminal value `Obsolete` ( $\rightarrow$  `Fixed.md`) with mandatory `**Obsolete-Details:**` audit-line (Since + Reason from a closed vocabulary + Superseding-item + Triple-check evidence). The §11.4.23 colorizer marks `Obsolete` rows light-gray + strikethrough. **Canonical authority:** constitution/Constitution.md §11.4.90.

### §11.4.91 — Summary-doc clarity

Inherited from constitution/Constitution.md §11.4.91. Every summary one-liner MUST be self-contained —  $\geq 6$  words OR  $\geq 40$  chars naming SUBJECT + PROBLEM/GOAL. Forbidden anti-patterns: section labels (`Composes with`, etc.), bare metadata (`Critical`, `Bug`, `In progress`), §-letter alone. Generators extract from the H1/H2 heading line per §11.4.54 ATM-NNN convention. **Canonical authority:** constitution/Constitution.md §11.4.91.

### §11.4.92 — Multi-pass change-evaluation discipline

Inherited from constitution/Constitution.md §11.4.92. Every non-trivial change passes 5 evaluation passes BEFORE commit: (1) main-task captured evidence; (2) regression blast radius; (3) cross-feature interaction; (4) deep research + CodeGraph queries; (5) anti-bluff confirmation. Trivial exemption applies only when zero source code is touched and the commit message cites it. **Canonical authority:** constitution/Constitution.md §11.4.92.

### **§11.4.93 — SQLite-backed single source of truth for workable items**

Inherited from constitution/Constitution.md §11.4.93. Text trackers (Issues / Fixed / Summary / CONTINUATION) become derived views of an authoritative SQLite DB at docs/workable\_items.db with a Go binary at cmd/workable-items/ providing bidirectional md-to-db / db-to-md sync. Round-trip must be byte-identical within tolerated whitespace. commit\_all.sh refuses to commit while the diff is non-empty. **Canonical authority:** constitution/Constitution.md §11.4.93.

### **§11.4.94 — Zero-idle priority-first parallel-by-default operating mode**

Inherited from constitution/Constitution.md §11.4.94. Idle is permissible ONLY when every queued item is genuinely externally blocked, OR operator STOP, OR §12 host-safety demand. Before any wake/sleep the conductor MUST survey parallel-work feasibility per §11.4.42 / §11.4.72 / §11.4.87 and dispatch non-contending items via §11.4.20 / §11.4.70 subagent-driven default. **Canonical authority:** constitution/Constitution.md §11.4.94.

### **§11.4.95 — Workable-items SQLite DB TRACKED in git, NEVER gitignored**

Inherited from constitution/Constitution.md §11.4.95. Amends §11.4.93: docs/workable\_items.db is authoritative source data, NOT a build artefact — TRACKED in git, never gitignored. Every md-to-db mutation stages + commits + pushes the DB alongside the MD regen per §11.4.19 atomic-move + §2.1 multi-upstream push. **Canonical authority:** constitution/Constitution.md §11.4.95.

### **§11.4.96 — Safe-parallel-work-with-long-build catalogue**

Inherited from constitution/Constitution.md §11.4.96. Classifies parallel work during long builds as SAFE (docs/scripts/tests/submodule pushes per §11.4.88, research, DB ops, subagent dispatch per §11.4.20) vs UNSAFE (destructive git ops on source tree, mass deletes on hot subtrees, host-safety breaches). Conductor consults the catalogue at every pause point. **Canonical authority:** constitution/Constitution.md §11.4.96.

### **§11.4.97 — Maximum-use-of-idle-time + progress-update cadence**

Inherited from constitution/Constitution.md §11.4.97. Strengthens §11.4.87 + §11.4.94 + §11.4.96: any idle minute during which work could autonomously progress and is not genuinely blocked is a §11.4.97 violation. Progress-update cadence: emit a 1-line operator-facing update at every commit landed / subagent return / anchor landing / captured-evidence acquisition / milestone closure. **Canonical authority:** constitution/Constitution.md §11.4.97.

## **§11.4.98 — Full-Automation Anti-Bluff: live tests re-runnable without manual intervention**

Inherited from constitution/Constitution.md §11.4.98. Closes the manual-intervention gap §11.4.85/§11.4.87/§11.4.89/§11.4.94 did not explicitly forbid: every test (unit / integration / e2e / Challenge / stress / chaos / live) MUST be fully self-driving end-to-end and emit PASS/FAIL/SKIP-with-reason without further human action after startup. One-time credential bootstrap OUTSIDE test execution is the only permissible exception. Re-runnability = PASS at `-count=3` consecutive automated invocations. **Canonical authority:** constitution/Constitution.md §11.4.98.

## **§11.4.99 — Latest-Source Documentation Cross-Reference Mandate**

Inherited from constitution/Constitution.md §11.4.99 (commit 9e3bcc5 ff 2026-05-28). Before committing any operator-facing instruction / guide / manual / setup walkthrough / troubleshooting cookbook, the author MUST fetch the LATEST official online docs of the third-party service / library being documented (WebFetch, MCP server, direct browsing — NOT training data) and cross-reference each instruction. A `## Sources verified <date>` section at the bottom of the document + a `Sources verified <date>: <urls>` footer in the commit message are MANDATORY. Re-verification cadence: every major release boundary;  $\leq 6$  months for general docs;  $\leq 90$  days for risk-classified families (messengers, cloud APIs, payments, LLM providers, code-hosting, OS/package managers). Stale docs graduate to §11.4.90 Obsolete after the 30-day grace. **Canonical authority:** constitution/Constitution.md §11.4.99 (commit 9e3bcc5). Forensic anchor: Herald MTPProto guide near-miss case study where stale guidance could have caused Telegram account ban.